GAME DESIGN DOCUMENT (GDD)

Game Name

Chainy. Chainy: A 3D World Educational Game for learning emerging Technology (Blockchain & AI)

Genre

Single-player 3D educational exploration and puzzle game.

Game Elements

Explore, interact, read short examples, answer Quiz questions, use Wordle clues, make BlockMatch moves, review explanations and save progress.

Player

One learner at a time; ages 12-17, especially underserved communities. Shared phones use separate local profiles.

TECHNICAL SPECS

Technical Form

Rounded 3D scenes and characters with 2D interface panels.

View

A camera presents the explorable 3D level. Exact angle, movement and camera tuning remain design decisions; activity panels focus attention on the active mini-game.

Platform

Android. Android 10+/11+, 4 GB physical RAM, 64 GB total device storage, ARM64, OpenGL ES 3.2 or Vulkan 1.1, entry-level-to-midrange GPU.

Language

Unity/C# for development; English for the first learning content.

22 October 2026 | Filled using the approved October 12 replacement design.

GAME PLAY

Device

Mobile touchscreen phone. Local SQLite stores profiles, content and progress. All required gameplay works offline.

You enter a small island learning area with the blue-haired explorer. A friendly robot asks you to inspect a community record and explains why people keep copies. You start a Quiz, choose an answer and read why it is correct or needs another look. In another activity, you guess a technology word and connect it to its definition. A BlockMatch board gives you a matching objective and explains the learning connection after the result. Only one mini-game is active at a time. After saving, you return to the level or leave; the same device can load your saved progress later.

Game Play Outline

Item	Chainy design
Opening the game application	Create/select a local learner profile, then start or continue.
Game options	Readable instructions, audio controls, pause and safe return.
Story synopsis	A welcoming technology learning community introduces everyday information problems.
Modes	Quiz, Wordle and BlockMatch; one active at a time.
Game elements	Explore, learn, act, receive feedback, revise and save.
Game levels	Compact explorable areas; exact count and sequence remain open.
Player's controls	Touch movement/interaction in the level; mode-specific controls in activities.
Winning	Meet the agreed activity objective; save before showing saved completion.
Losing	Wrong answers, exhausted guesses or unmet board objectives follow agreed retry rules.
End	Read the result and explanation; continue, retry or exit.
Why is all this fun?	Exploration, clues and visible feedback offer achievable challenges; enjoyment needs testing.

DESIGN DOCUMENT

Key Features

Offline access; approachable 3D characters; three varied mini-games; plain-language explanations; shared-device profiles; saved local progress.

Design Guidelines

An offline Android educational game introducing foundational Web3, Blockchain and AI to learners aged 12-17, especially learners in low-income and remote communities with limited technology access.

Preserve the approved rounded 3D visual style. Essential feedback works without audio and never relies only on color. Offer practice for unfamiliar touch controls. No real cryptocurrency, external authentication or cloud AI is required.

Game Design Definitions

Reference	Object, property and action
1 Menu / profile	Selected learner and saved point; choose, continue or switch.
2 Level / synopsis	Explorer position and learning objective; move, interact and read.
3 Active mini-game	Question, guess or board state; answer, submit or make a move.
4 Result / game over	Outcome and explanation; review, retry or return.
5 Progress / save	Local saved state; commit, retry save or resume later.

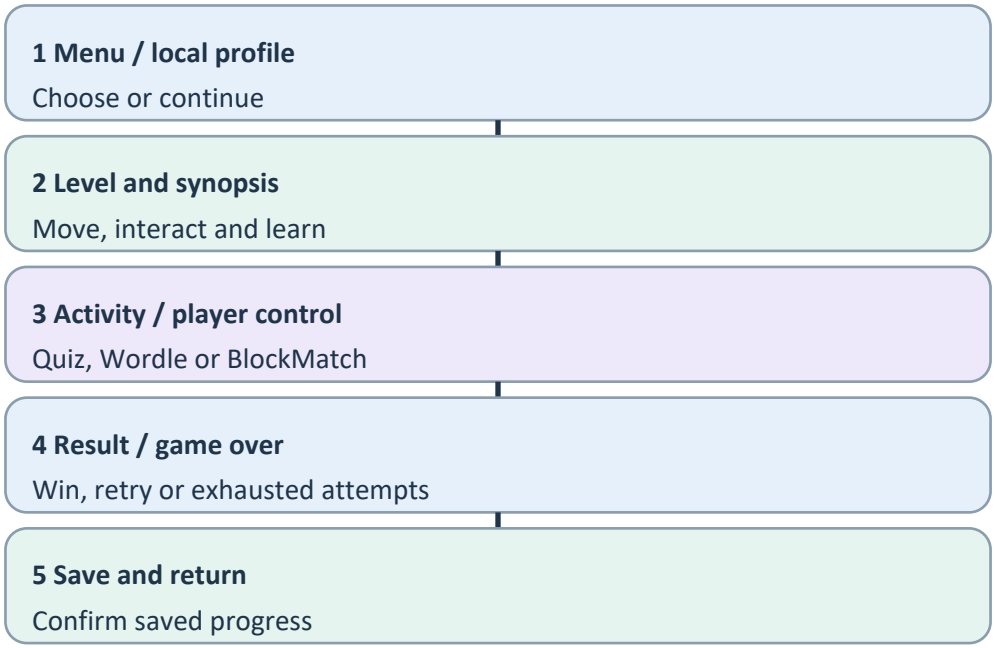
Winning, losing and transitions

Only one activity receives input. After its outcome, the learner reviews an explanation and follows the permitted next action. Completion/pass values, Wordle limits and BlockMatch endings are still open. Save failures preserve earlier progress and show retry. The result is not a combat death state.

These references identify the elements in the flowchart on the next page. They are document references, not software class names.

Game Flowchart

The numbers match Game Design Definitions on page 3. The flow shows the objects, their changing state and the player actions between them.



Save failure: keep earlier progress and offer retry. Pause: resume or leave safely.

Menu and synopsis

1 selects the local profile. 2 introduces the level and learning objective before practice.

Game play and player control

3 routes input only to Quiz, Wordle or BlockMatch. Movement resumes when the activity is closed. Pause and loading must prevent accidental submissions.

Game over: winning and losing

4 explains success, a mistake or an exhausted attempt allowance under the agreed mode rules. The learner retries where allowed or returns safely. 5 confirms saving before showing saved completion.

Recovery

An interrupted session resumes from its last valid saved point. A learner code on another device does not recover an old save. Teacher file transfer and app installation/updates are separate from offline gameplay.

Player Definition

The learner controls the explorer and the active activity. The character is a friendly guide into technology learning, with no combat role.

Player Definitions

Health: not applicable; no health depletion is required. Weapons: not applicable. Actions: move, interact, read, select, answer, guess, make a board move, pause, review and retry.



Proposed character interpretation; use the original approved style references for final art.

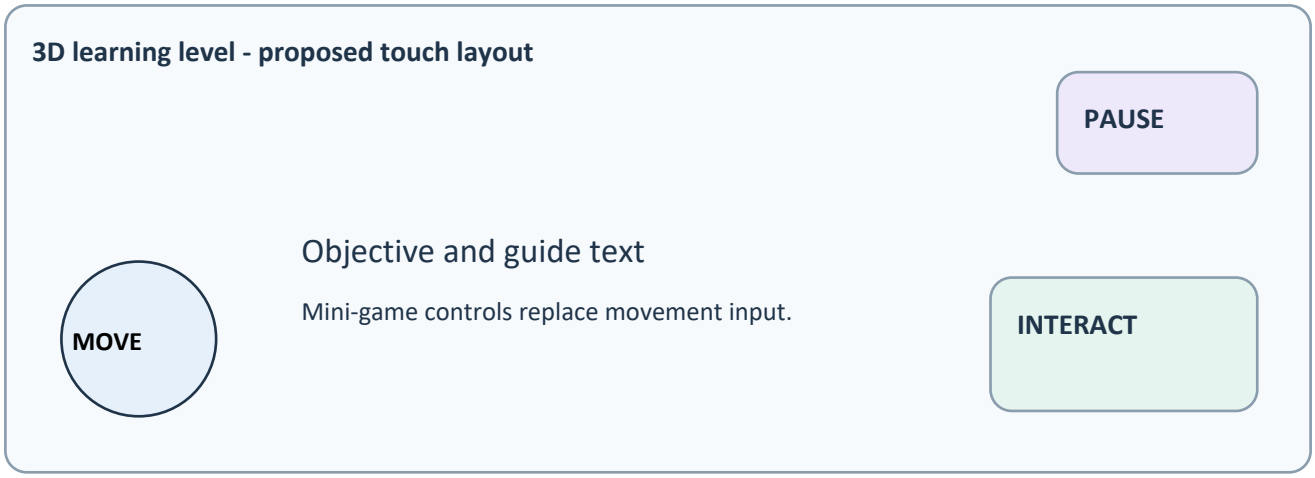
Player Properties

Property	Visible feedback
Active learner	Profile label changes; previous learner data is cleared from the screen.
Position / interaction	Explorer moves; a nearby interaction becomes available.
Active mini-game	Only its controls and current objective receive gameplay input.
Answer, guess or board	Selection, letter feedback or match result reflects the action.
Result and saved progress	Outcome is explained; saved confirmation appears only after a successful write.

Player Rewards (power-ups and pick-ups)

Positive feedback explains success and saved progress makes achievement visible. Exact unlocks or badge rewards remain design decisions. No health pickup, paid reward, currency economy or power-up system is required. Keep explanations available after the celebration.

User Interface (UI)



Proposed arrangement for touch testing. It does not assert the final camera or control layout.

Controls and screen behavior

Use a movement area and context interaction button in the 3D level. During Quiz, show readable answer choices and a clear submission state. During Wordle, show guess input and letter feedback with labels or symbols. During BlockMatch, show the objective and legal board interaction; wait for cascades before accepting another move.

Readability and recovery

Keep objective, Pause, instructions and result feedback visible. Use large touch targets, readable text and audio-independent explanations. Show save errors with a clear retry. On shared devices, switching learners clears the previous profile from the screen.

Rules and ownership

Sherwin decides the learner experience; Christian supplies implementation facts; Luis owns art; Sean authors documents; Miguel implements UI. Quiz, Wordle and BlockMatch exact rules must be agreed before production content is frozen.

Delivery context and sources

Fixed beta: November 3-6, 2026, Tambunan National High School, Marinduque. Fixed pre-final: November 21, 2026. These dates do not prove readiness. Sources: supplied six-page GDD template; Christian's adopted technical design; current project baseline and game design. The supplied rubric totals 32 points; the school completion calculation remains unresolved.

Original template field order retained. Instructional filler is replaced with project content; no external template instruction authorizes extra features.

Visual supplement A - exploration

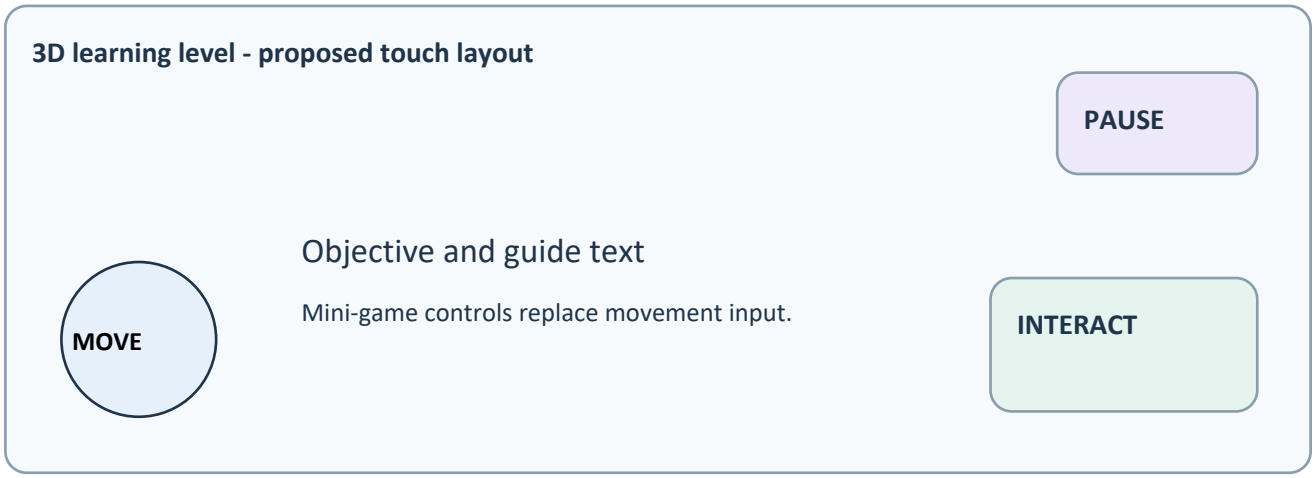
This plate accompanies Game Play, Design Guidelines and User Interface. It applies the approved character proportions, yellow costume, friendly robots and rounded 3D island forms to the current exploration design.



Figure A. Generated concept: explorer, guide and learning station. Environment size, camera and props are illustrative, not final level requirements.

How the learner reads this scene

The path supports movement toward a visible guide. The book and data cube suggest a learning interaction. Dialogue and activity instructions should appear in a clear panel without covering the important object.



The control schematic shows proposed touch positions; the illustration above supplies visual direction. Test the final layout on target phones.

Visual supplement B - mini-games

This plate accompanies Modes, Game Design Definitions and UI. The three panels are separate activity concepts, with the approved character style carried through each.

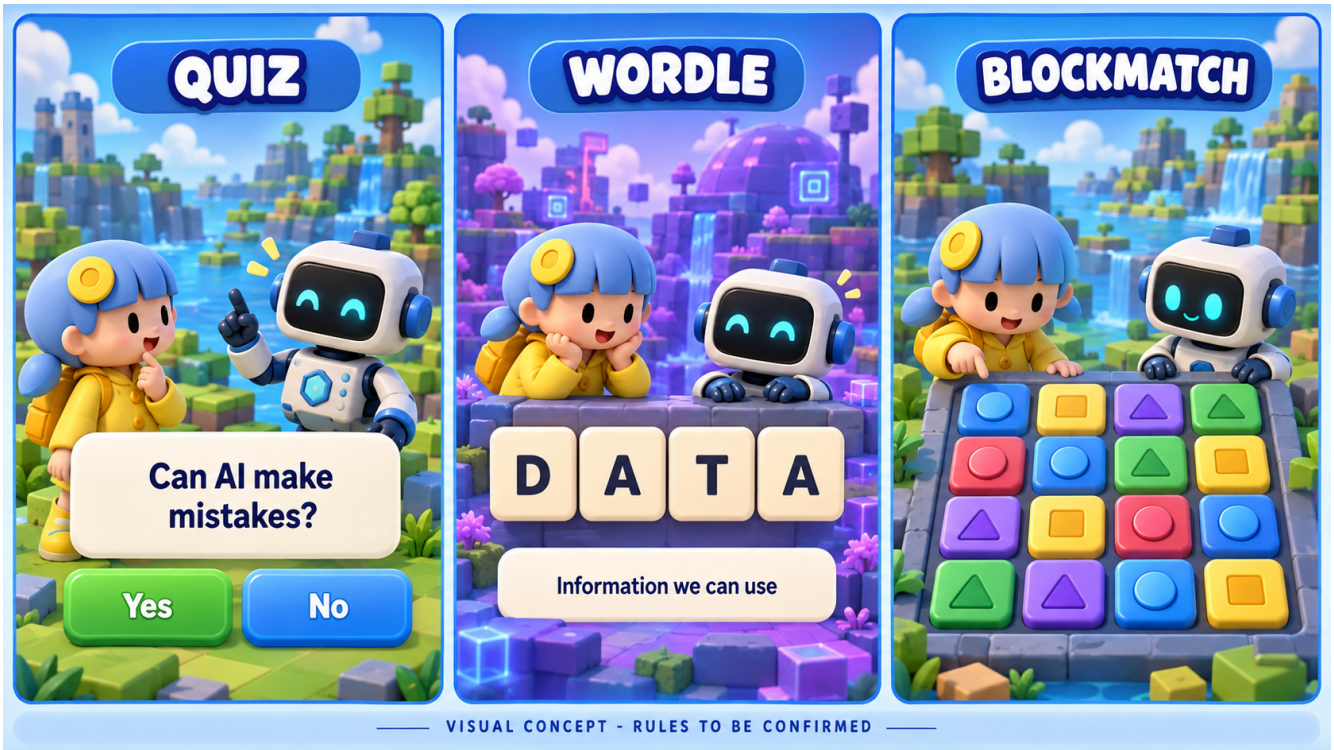


Figure B. Generated concept sheet. Shown word length, tile count and colors do not approve rules. Only one mini-game runs at a time.

Quiz: choose and understand

The sample asks whether AI can make mistakes. The correct answer is Yes. The final activity should explain that AI answers need checking; button color must not reveal the answer before submission.

Wordle: connect words to meaning

DATA is a vocabulary example. It does not fix word length, allowed guesses or repeated-letter rules. Pair the agreed letter feedback with an explanation of the word.

BlockMatch: distinguish tiles

Symbols distinguish tiles alongside color. Board dimensions, matching rules and the educational connection require approval. This concept does not show a legal move or winning state.

Both plates were generated with the built-in image tool using reference-chainy-01.png and reference-chainy-02.png as style references. Prompts and provenance are recorded in docs/media/gdd/current-visuals/README.md.